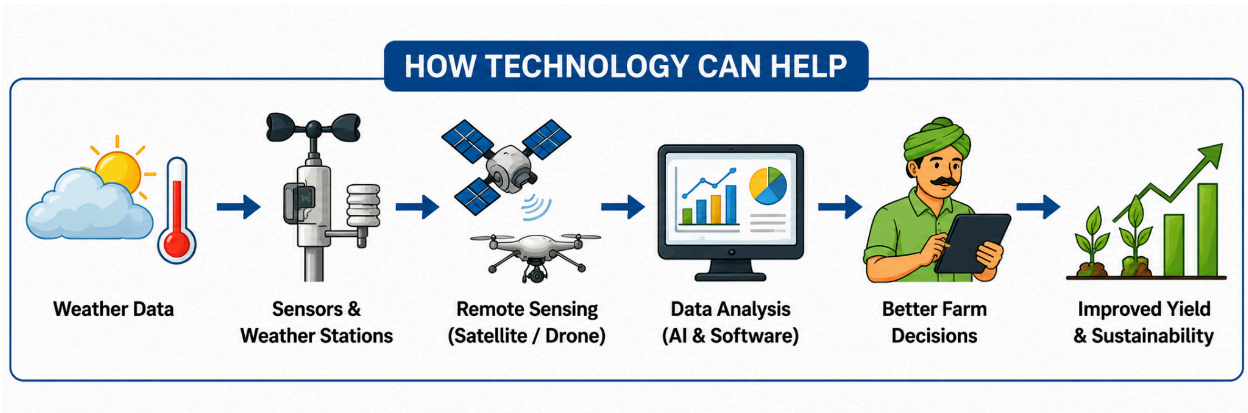
UNIT - III
AGRICULTURAL SYSTEM MANAGEMENTS

Q1. Challenges Faced in Agriculture System Under Climate Conditions

Climate conditions have a major effect on agriculture. **Temperature, rainfall, drought, floods, changing weather patterns, pests and diseases** can directly affect crop production.



1. Unpredictable Weather

Weather patterns are becoming difficult to predict.

Problems:

- Sudden changes in temperature.
- Unexpected rainfall.
- Changes in the farming season.

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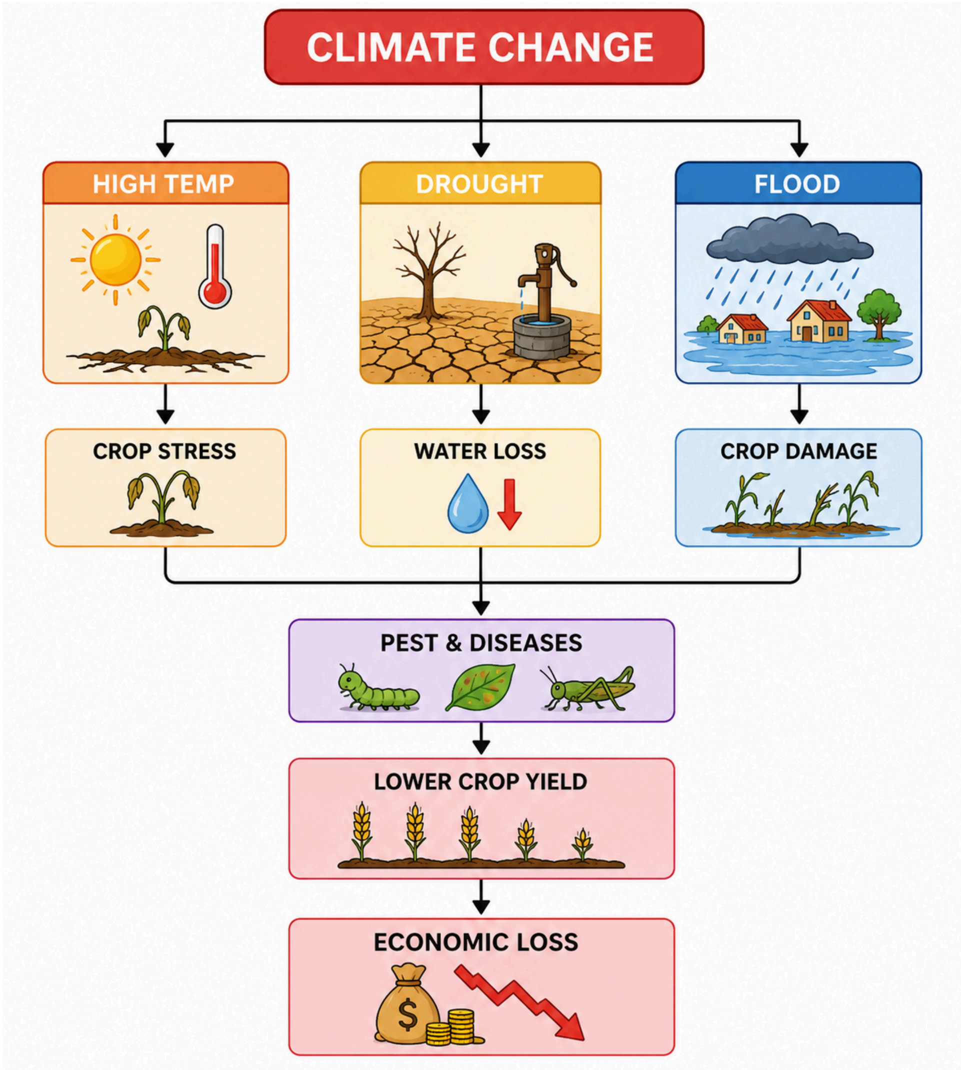
2. High Temperature and Heat Stress

Very high temperature can affect crop growth.

Problems:

- Plants may lose more water.
- Crop growth can become weak.
- Flowering and fruit formation may be affected.



3. Drought and Water Scarcity

Drought means a long period with **very little rainfall**.

Problems:

- Less water is available for irrigation.
- Soil moisture decreases.
- Crops may dry up.
- Crop production is reduced.

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4. Difficulty in Crop Selection

Different crops require different temperatures, rainfall and soil conditions.

Problems:

- A crop suitable for one season may not perform well in another.
- Changing climate can make traditional crop choices less suitable.
- Farmers need to select suitable crop varieties.

5. Increase in Pests and Diseases

Changes in temperature and humidity can create suitable conditions for **pests and diseases**.

Problems:

- More pest attacks.
- Faster spread of diseases.
- Crop damage.
- Increased pesticide requirements.

6. Economic Loss

Climate problems can increase the cost of farming.

Problems:

- Higher irrigation cost.
- More spending on pesticides.
- Crop losses.
- Reduced income.
- Higher cost of inputs.

Q2. Artificial Intelligence System and Decision Support System in Agriculture

Artificial Intelligence in Agriculture

AI is becoming important in agriculture because farming depends on many changing factors such as **soil, weather, moisture, crop condition, and temperature**.

Main uses of AI:

1. Weather Prediction

AI can analyze weather information and help predict Rainfall, Temperature, Weather changes. This helps farmers plan **planting, irrigation, and harvesting**.

2. Soil and Crop Health

AI can analyze soil and crop data to identify: Soil condition, Crop stress and Poor crop growth. This helps farmers provide the required **water and nutrients**.

3. Pest and Disease Detection

AI can analyze images from **drones and satellites** to identify: Pests, Plant diseases, Unhealthy crops

4. Yield Prediction

AI and machine learning can use **historical data and current agricultural data** to predict crop yield.

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2. Decision Support System (DSS)

A **Decision Support System** acts as a bridge between **agricultural data and human decision making**. It converts collected information into useful recommendations for farmers and managers.

Main functions of DSS:

1. Data Integration

DSS can combine data from different sources such as Weather data, Soil data, Crop data, Market data

2. Data Analysis

DSS analyzes large amounts of agricultural data and identifies useful patterns.

3. Risk Management

DSS helps identify risks related to: Weather, Pests and Diseases This helps farmers take preventive action.

4. Scenario Analysis

DSS allows farmers to evaluate different choices.

For example:

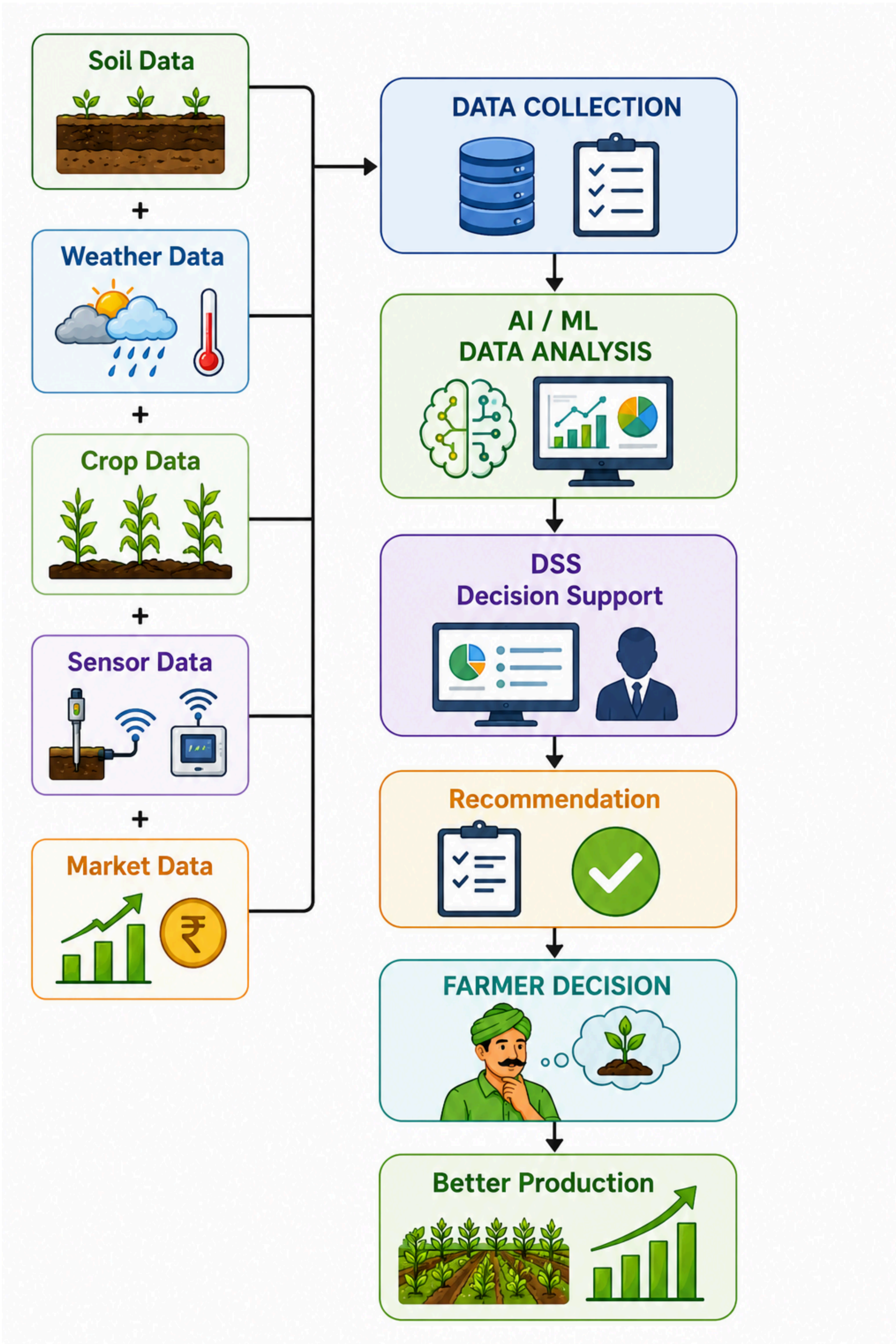
Crop A → Low water requirement
Crop B → High water requirement

5. Resource Allocation

DSS helps farmers use available resources properly.

Resources include: Land, Water, Fertilizers, Labour and Money

AI + DSS



Soil Data
+
Weather Data
+
Crop Data
+
Sensor Data
+
Market Data
DATA COLLECTION
AI / ML
DATA ANALYSIS
DSS
Decision Support
Recommendation
FARMER DECISION
Better Production